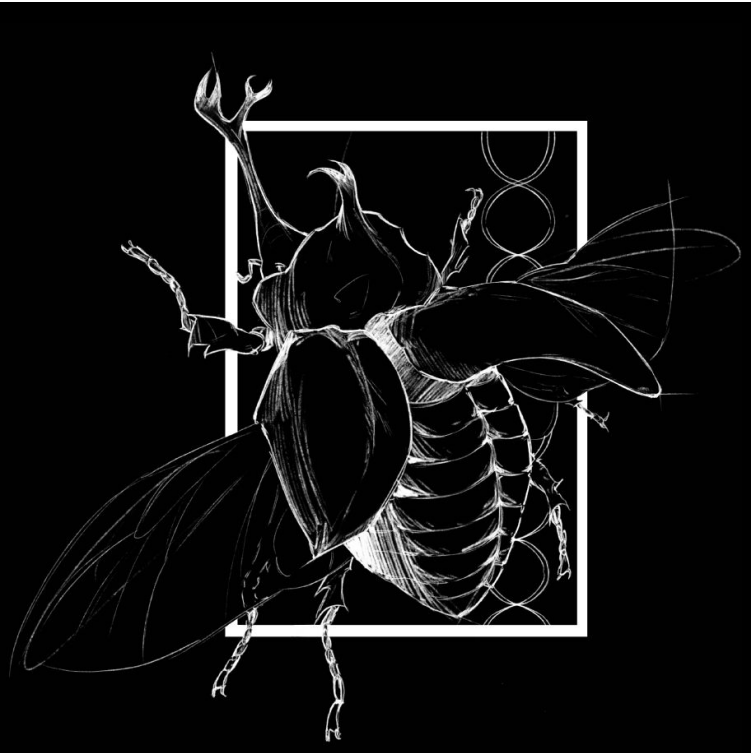




The Influence of Island and Mainland Environments on Chromosome Number Evolution in Serpentes

Anaya Hooda and Heath Blackmon

Department of Biology, Texas A&M University, College Station, TX 77843



Background

Chromosome number evolution is an important driver of genomic diversity and may play a role in speciation across vertebrates, including snakes. In snakes, karyotype variation is largely shaped by chromosomal fissions and fusions, yet the evolutionary forces influencing how rapidly these changes occur remain poorly understood.

The Drift Barrier Hypothesis provides a framework for understanding variation in evolutionary rates. This hypothesis proposes that the power of natural selection is limited by genetic drift, particularly in populations with small effective population sizes (N_e). In such populations, slightly deleterious mutations, including chromosomal rearrangements, can accumulate more readily because selection is less effective at removing them.

Island systems offer a natural setting to test this hypothesis. Species classified as island endemic, those restricted to isolated island environments, often have smaller population sizes and reduced gene flow compared to mainland species. These conditions increase the influence of genetic drift, potentially accelerating the accumulation of genomic changes over time.

In this study, we test whether island endemic snake species exhibit faster rates of chromosome number evolution than mainland species. Using haploid chromosome number data mapped onto a phylogenetic framework, we apply chromosome evolution modeling and simulation-based approaches to compare rates of chromosomal change between island and mainland lineages.

Study Design & Model Fitting

Haploid chromosome numbers came from a vetted dataset provided by my professor. I standardized species names against currently accepted usage, removed duplicates, and dropped outliers flagged in the original notes. The cleaned list was matched to a time-calibrated snake phylogeny; taxa without a one-to-one match or with unresolved karyotypes were excluded. This produced a tree with tip states for n and, after geographic coding, a regime label per species.

Geographic classes combined a published distribution dataset (island and continental polygons with areas in square meters) with manual checks in the Reptile Database. A species was coded as island endemic only if all mapped occurrences fell within island polygons; continental, mixed, and broadly distributed species were coded as mainland. Ambiguous cases were resolved conservatively, defaulting to mainland unless island endemism was clear. Species were written to a CSV pairing haploid number with island/mainland class, then joined to the phylogeny.

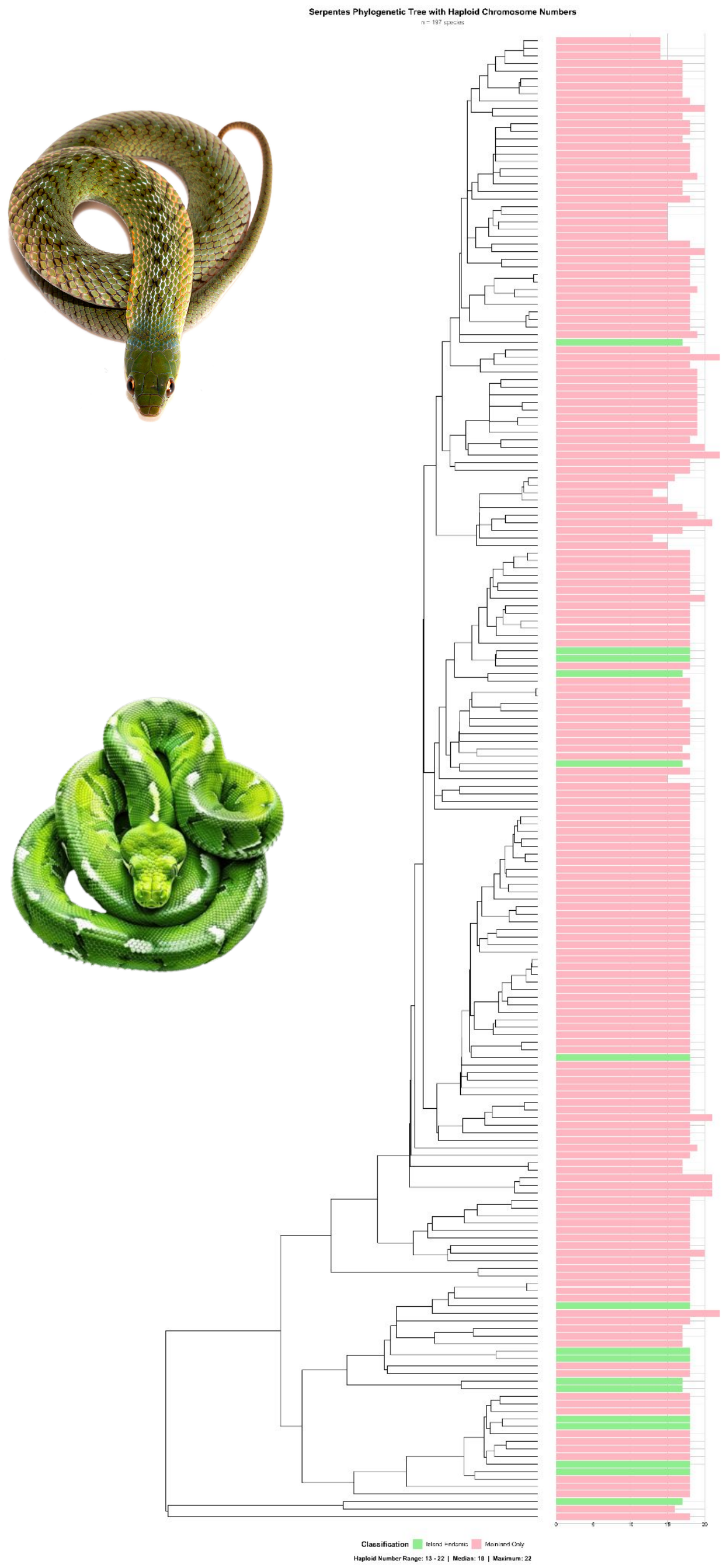
Analyses used ChromePlus in R, which models chromosome number as a continuous-time Markov process with three event types: ascending steps ($n \rightarrow n+1$, fissions), descending steps ($n \rightarrow n-1$, fusions), and doubling ($2n$, polyploidy). The model estimated separate rates for each process in each regime, plus low regime-transition rates along branches to accommodate historical range changes. Priors and MCMC settings followed ChromePlus vertebrate defaults, with chains run long enough for stable posterior summaries.

Model checking focused on convergence. I inspected trace plots, verified adequate effective sample sizes, and confirmed that independent chains converged to the same stationary distribution. Posteriors were visualized as (i) a phylogeny with island tips in green and mainland tips in pink, and (ii) densities of Mainland – Island rate differences for each process. Inference rests on posterior overlap across regimes rather than separate null simulations.



AI Use in Project

Artificial intelligence tools were used to assist with coding, data processing, and figure generation. AI helped generate and refine R scripts used for species classification (island vs. mainland), data organization, and visualization. All code was reviewed, tested, and executed by the researcher. Outputs were validated to ensure accuracy prior to inclusion in the analysis.



Rate Differences: Mainland vs. Island

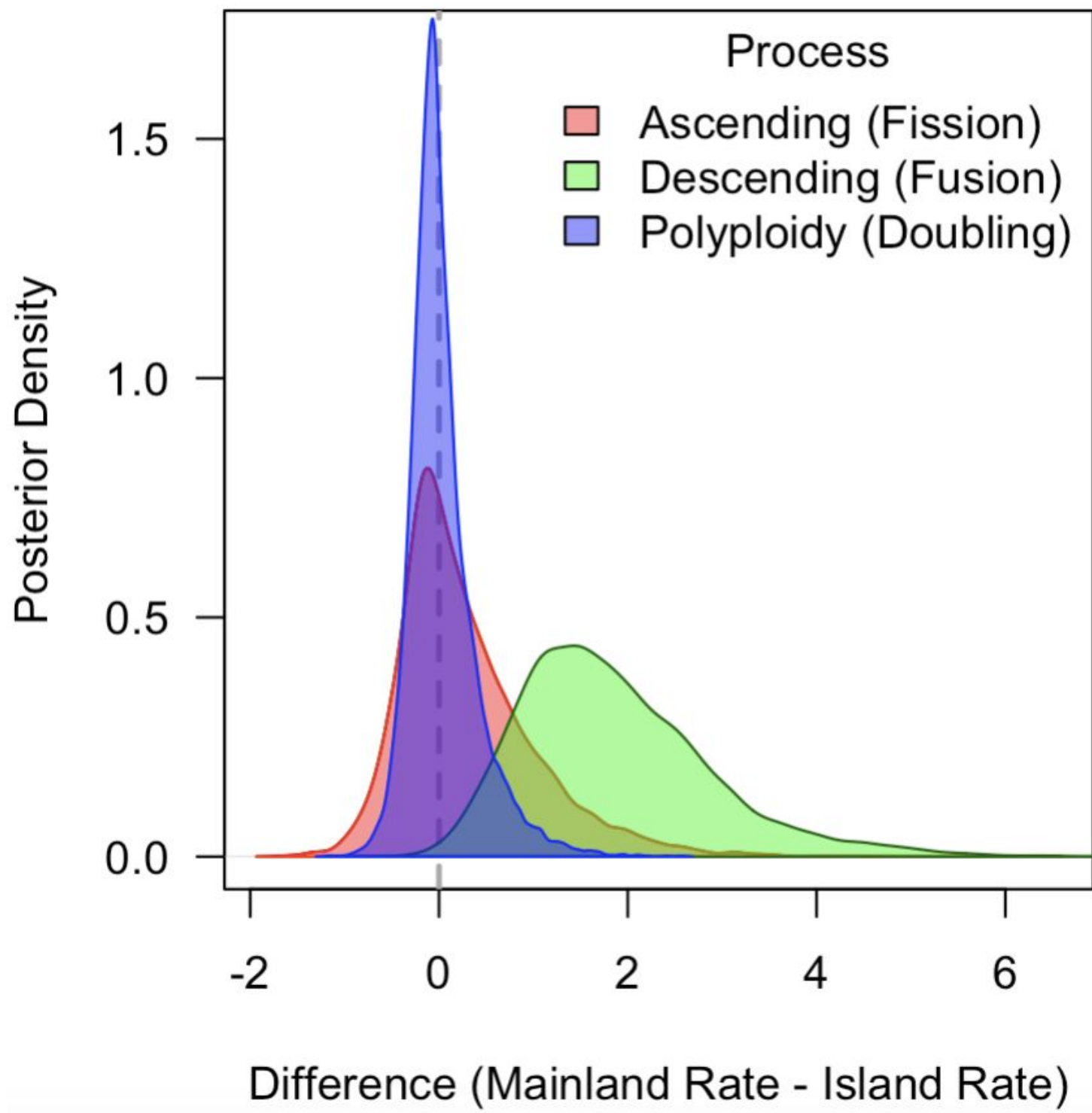


Figure 1: Time-scaled snake phylogeny with tips colored by geographic class—green for island endemics and pink for mainland/widespread species. Island taxa are scattered across the tree rather than confined to a single clade, illustrating multiple independent origins of island endemism and providing broad phylogenetic coverage for the regime comparison.

Figure 2: Posterior density of rate differences (Mainland – Island) for the three processes. Red curve = fission, green = fusion, blue = polyploidy. The vertical dashed line marks zero (no difference). Fission and polyploidy curves are centered at zero with tight overlap, indicating no regime effect. The fusion curve is shifted slightly to the right but still overlaps zero substantially, reflecting a weak, uncertain mainland tilt rather than a credible difference.

Discussion

Our results do not support the idea that island-only snake species change chromosome number faster than mainland or widespread species. Even though islands often have small populations where chance can push changes to fixation (genetic drift), the estimated island rates closely match mainland rates for the key processes that change chromosome count. In the rate-difference plot, the fission curve is centered at essentially zero with broad overlap across regimes, and the polyploidy curve collapses near zero for both—consistent with the biology of snakes where whole-genome doubling is effectively absent. The fusion curve tilts slightly to the right (suggesting higher mainland rates), but its uncertainty still spans zero, so there is no credible island acceleration.

Many chromosome-number changes reduce fertility because the altered chromosome can't pair correctly with its normal partner during meiosis. As a result, carriers of new fusions or fissions leave fewer viable offspring, so these changes rarely spread—on islands or on the mainland. Those costs depress fixation probabilities even when drift is stronger, which helps explain why island and mainland rates look similar. Reptiles also have a highly conserved chromosome architecture—including microchromosomes and ancient synteny—that limits which rearrangements are even viable. Under this constraint-dominated scenario, similar rates across geography are what we should expect unless a lineage's recombination or centromere biology changes in a way that relaxes those constraints.

The slight hint of higher fusion on the mainland can be understood without invoking a real biological difference. Sampling is imbalanced, there are many more mainland taxa, so their rate estimates are tighter, while island posteriors are wider. Many islands are geologically young or have recent colonists, leaving less time for number changes to accumulate. The mainland also spans a broader set of environments and demographic histories, which can occasionally favor fixation of certain fusions (for example, if they alter recombination in ways that hitchhike with adaptation). However, because the credible intervals around the fusion difference overlap zero substantially, these ideas remain plausible explanations for a weak, noisy pattern rather than evidence of a true mainland–island contrast.

Essentially, islands do not generally speed up changes in chromosome number in snakes. If island and mainland snakes differ, those differences likely come from other processes, ecological adaptation, behavior, patterns of gene flow, or genome rearrangements that shuffle DNA without changing the number of chromosomes (for example, inversions or exchanges between micro- and macrochromosomes that keep n the same). To test this more rigorously, future work should replace the simple island/not-island label with quantitative island traits (area, isolation, age, and time since colonization), use simulations on the actual phylogeny to see how large an island–mainland difference could appear just by chance, and broaden the analysis beyond chromosome number to track inversion rates, micro-to-macro fusions, and sex-chromosome evolution. These steps will clarify where geography truly leaves a genomic signature—and where it does not.

Conclusions & Acknowledgements

Roll, U., & Meiri, S. (2022). GARD 1.7 – Updated global distributions for all terrestrial reptiles (Version 1.7) [Data set]. Dryad. <https://doi.org/10.5061/dryad.9cnp5hqmb>
Román-Palacios, C., Medina, C. A., Zhan, S. H., & Barker, M. S. (2021). Animal chromosome counts reveal a similar range of chromosome numbers but with less polyploidy in animals compared to flowering plants. *Journal of Evolutionary Biology*, 34(8), 1333–1339. <https://doi.org/10.1111/jeb.13854>
Kumar, S., Suleski, M., Craig, J. M., Kaspróicz, A. E., Sanderford, M., Li, M., Stecher, G., & Hedges, S. B. (2022). TimeTree 5: An expanded resource for species divergence times. *Molecular Biology and Evolution*, 39(8), msac174. <https://doi.org/10.1093/molbev/msac174>
Uetz, P., Freed, P., Aguilar, R., & Hošek, J. (Eds.). (2026). *The Reptile Database*. Retrieved April 21, 2026, from <http://www.reptile-database.org>
Acknowledgements: Thanks to Kenzie Laird and the Blackmon Lab for guidance on chromosome number modeling and ChromePlus workflows.